

Attention

Minister of Construction, Mr. Tran Hong Minh

Title

Anticipatory Analysis for Strategic Transport Policy

Purpose

This briefing outlines how far-sighted anticipatory analysis can inform ambitious, health-conscious and economically sound transport policy decision-making in Viet Nam.

Executive Summary

Strategic transport planning facilitates sustainable economic growth while improving health outcomes. Viet Nam has the opportunity to increase its planning ambition, moving from reactive short-term planning towards a long-term view that integrates the country's strategic objectives.

An analytical framework is explored to support decision-making. Establishing a long-term vision through Backcasting anchors policy choices. Forecasting tools are then examined to align this vision with socio-economic trends. The framework supports financial considerations by increasing confidence, as well as identifying health externalities and strategic synergies. A cyclical framework is proposed for implementation to facilitate strategy evolution. Table 1 summarises recommended tools.

Table 1: Recommended Anticipatory Toolkit

Tool	Description	Decision Support
Backcasting	Develop an ambitious vision of future Viet Nam.	Anchor policy decisions to long-term strategy.
Forecasting	Qualitative and Quantitative techniques, including Pathway Analysis, Qualitative Comparative Analysis, and Econometric modelling using a digital twin.	Identifies trends shaping transport needs and builds confidence in decisions.
Synergy Identification	Analyse overlap and complementarities between transport policy and other development goals.	Integrates transport policy with wider strategic objectives.
Externality Identification	Quantify spillovers from transport investment that are omitted from traditional financial modelling.	Explicitly incorporates health outcomes into decision-making
Iterative Development	Define a policy reporting and update cycle.	Ensure policy responsiveness and strategic evolution.

Problem Context

Designing Viet Nam's transport policy while our economy grows rapidly is challenging. Considering the health outcomes linked to transport sector emissions adds complexity. Anticipatory analysis will move us beyond reactive policy, creating a transport system that encourages economic growth without compromising health.

A decade of rapid growth, summarised in Table 2, has strained our transport system. Manufacturing exports are driving GDP growth, pressuring transport corridors.

	2014	2024
GDP per Capita	\$2559	\$4700
FDI Inflows	\$9.2 billion	\$20.2 billion
Poverty Rate	8%	2%
Gini Index of Inequality	0.43	0.37
Population	91 million	101 million
Urban Population	32%	38%

Table 2: Data Sources: *National Statistics Office of Viet Nam, World Bank Indicators*. Gross Domestic Product (GDP) and Foreign Direct Investment (FDI) quoted in current USD.

This growth has come at the expense of air quality. Fine Particulate Matter (PM_{2.5}) emissions affect health directly and have been rising as our economy grows. Figure 1 shows an increasing portion of emissions attributable to domestic navigation, as our exports increase. Transport sector CO₂ emissions have also been rising 2% annually since 2015. Viet Nam is heavily reliant on road transport, accounting for 84% of domestic transport sector CO₂ emissions in 2022. Two-wheel vehicles account for 94% of the vehicle fleet, and low rates of urban rapid transport provide little alternative (World Bank Group, 2022b). There is scope for optimism, however, as the country's first subway system has opened in Ho Chi Minh City. Efficiency improvements have also driven a 50% reduction in emissions intensity of GDP since 2000, but this is outweighed by increased economic activity (Asian Transport Observatory, 2024). In consequence, Viet Nam has the 22nd-worst air quality globally. Hanoi, in particular, ranks as the 8th most polluted city in the world (Pratt et al., 2024).

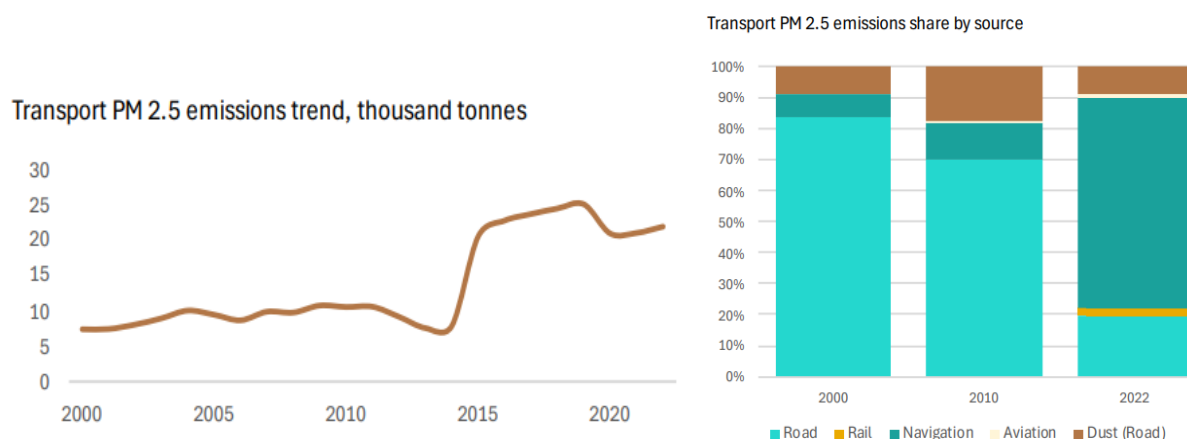


Figure 1: Source: Asian Transport Outlook (Navigation and aviation only includes domestic transportation)

The health outcomes linked to these statistics are very real. In 2017, there were over 70,000 deaths attributed to the poor air quality (Health Effects Institute, 2024). It is estimated that 7% of total deaths in Viet Nam are attributable to air pollution.

The National Action Plan on Pollution Remediation and Air Quality Management 2026-2030 outlines our current efforts to tackle emissions (Government of Vietnam, 2025). The plan is ambitious, costing approximately €4.6 billion over five years. A 20% reduction in PM2.5 concentration by 2030 is targeted for Hanoi, with similar targets across the country.

Evident even from its title, the plan largely focuses on defensive actions aimed at building regulatory and monitoring capabilities. Technical standards and fuel quality monitoring are proposed for the vehicle fleet. Water sprinkler systems and green spaces are planned to help clear the air of harmful pollutants. While these measures are undoubtedly beneficial, they fail to address emissions at source. Notably, details on how the plan will be financed are omitted. Official development assistance has decreased as the country develops, creating a potential funding gap.

Our reactive stance poses a significant challenge, as continued economic growth may outweigh any improvements. Additionally, Viet Nam's transport system is also exposed to climate risk due to its low-lying river deltas and tropical climate. Road failures due to river flooding are expected to increase by at least 40% by 2030 (World Bank Group, 2025). The cost of repairs may divert funding from more strategic projects unless planned for.

Air pollution must be incorporated into transport policy, as it is the biggest source of pollution. The action plan makes little reference to a long-term transport vision in Viet Nam, and this brief seeks to address this shortcoming.

Backcasting

Transport planning is challenging because it interacts endogenously with all facets of the economy. Without a long-term strategic vision of the transport system, decisions become constrained by socioeconomic realities. Clarifying a future vision for Viet Nam is the first step towards strategic transport planning.

Backcasting is a planning approach that begins by defining a desirable future vision. The approach is useful for transport policy, where complex interactions make any attempt to forecast far from the present difficult (Dreborg, 1996). Instead, a long-term view is taken, beginning with the end-point and working backwards to identify how to achieve it.

Our National Green Growth Strategy 2021-2030 is a good starting point and contains high-level goals for 2050 (Government of Vietnam, 2021). The strategy identifies economic prosperity, environmental sustainability, and social equality as guiding principles for Viet Nam's future.

For the purpose of Backcasting, a complete and ambitious vision of Viet Nam in 2050 must be defined to anchor policy choices. Developing this vision requires multi-stakeholder consultation and cross-departmental alignment. Comparative analysis

provides a useful source of inspiration for our vision. The highly connected clean transport systems of South Korea and Japan are ambitious examples that may guide a Backcasting approach. Building on elements of the Green Growth Strategy and drawing from international comparisons, the basis for a vision might look like this:

2050 Strategic Vision of Viet Nam

- Viet Nam is now Highly Urbanised, driven by the growth of our digital economy. Our cities are competitive and attractive, drawing international talent.
- The transport system is environmentally resilient, with high-quality transport corridors between advanced manufacturing industry hubs and ports. Major cities are connected, and Urban Rapid Transport has largely displaced private vehicles.
- Transport emissions-related illness and death have been largely eliminated through fuel standards, electrification, and displacement by clean public transport.

Moving backwards from this vision allows decision-makers to identify high-level strategic considerations for transport policy. For example, the vision above might inform three strategic priorities guiding long-term policy decisions:

1. **Avoid** emissions through electrification and urbanisation.
2. **Shift** urban transport from private vehicles to integrated public transport.
3. **Improve** connectivity along freight corridors and between major cities.

Backcasting generates an ambitious vision of Viet Nam's future that guides high-level decision-making but may neglect current trends. Forecasting methods are often useful here to clarify what is obtainable.

Forecasting

Looking forward from today and predicting the consequences of potential transport and air quality policies is the domain of forecasting. Forecasting methods can be broadly grouped into qualitative and quantitative approaches, and their strategic application to transport is further outlined in this section.

Qualitative Methods

Pathway or Scenario Analysis is a common approach to forecasting complex problems where quantitative prediction falls short (Lyons & Chatterjee, 2002). In this case, we identify a range of possible themes that may shape our future. The objective here is to bound a more precise analysis by overarching macroeconomic, environmental, and demographic trends. Decision makers can refer to these scenarios as time passes, adjusting plans according to which scenarios are becoming more likely.

Typically, high and low scenarios are defined across economic, urbanisation, and environmental dimensions. Quantitative analysis, discussed further below, may help to clarify these scenarios, but the process is largely qualitative and exploratory. The choice of economic scenarios has important implications for air quality policy, given our reliance on road transport to support our export-driven economy. Developing a likely range of scenarios necessitates consultation with a variety of stakeholders

across government departments and industry (Pherson, 2018). Joint consideration of our Industrial and Land Use policies will help determine how the country is likely to develop.

Building a range of likely scenarios shapes our understanding of where and when transport investment is likely to be needed. For example, Backcasting identified our vision for a Digital Economy, but it is unclear how fast this will develop in response to global macroeconomic trends. Establishing high and low scenarios here will help policymakers strike a balance between rural and urban transport investment. Consideration of misalignments between the Backcasting vision of Viet Nam and these forecasted scenarios may highlight areas of weakness in both, thereby informing a more robust strategy overall.

Qualitative Comparative Analysis (QCA) can help establish where investment is likely to have the highest impact once bounds have been established through scenario analysis (Verweij & Gerrits, 2013). QCA is a useful framework for evaluating past projects influenced by complex policy systems. The method controls for contextual factors, allowing us to draw generalised conclusions about project performance. Health outcomes should be incorporated into the performance criteria. Conclusions can be used to decide on the most impactful modes of transport, considering regional characteristics.

The Qualitative techniques outlined above inform a clearer picture of where transport policy should focus. However, investment decisions typically rely heavily on quantitative analysis, and techniques for doing so strategically are explored in the following section.

Quantitative Methods

Developing a Digital Twin of Viet Nam's transport system can help to identify system needs ahead of time. Econometric Modelling using the digital twin integrates modal choices, emissions data, population growth and land use data to identify investment needs (Atkins, 1987). Models typically rely on statistical techniques including regression and more recently Monte Carlo simulation, to gauge the uncertainty of outcomes. A model of PM2.5 concentration will identify pollution hotspots under different modal shift scenarios, supporting health-conscious decisions. This kind of modelling is key to developing investment-grade decision-making. Public Private Partnerships are increasingly used to deliver transport infrastructure in Viet Nam, and the long-term nature of these contracts requires significant quantitative forecasting to manage risk.

Stress-testing plays an important role in transport modelling and strategy identification. Modelling outcomes in complex non-linear systems can be sensitive to input assumptions (Misina & Tessier, 2008). Feeding back into earlier stages of analysis, stress-testing is useful for identifying issues with the assumptions underpinning the Backcasting and Pathway Analysis. The overall process must be iterative, repeatedly updating and refining assumptions and pathways before decisions are made.

Financial Considerations

Viet Nam's economy is expanding rapidly, but financial constraints remain a central component of transport and health policy decision-making. The absence of financial detail within the National Action Plan on Pollution Remediation and Air Quality Management is therefore concerning. Anticipatory analysis strengthens the justification of transport and air quality investment by quantifying externalities and endogenous effects. Additionally, the analysis optimises investment within fiscal constraints by identifying strategic synergies.

Externalities

It is difficult to assess the benefits of transport projects without considering their wider impact on health and the economy. Identification of positive externalities provides an important justification for projects (Shiftan et al., 2002). The World Bank estimates that the health and social effects of air pollution translate into an annual cost of 13.3 billion USD (World Bank Group, 2022a). Anticipatory analysis can help quantify the reduction in this figure attributed to clean transport projects. This can inform Value for Money assessments that anticipate the true benefit of strategic transport projects.

Anticipatory forecasts using the digital twin can also identify where transport imposes a binding constraint on growth. Simulation methods can be used to construct a synthetic counterfactual without these endogenous effects. The difference in economic output can be attributed to the proposed transport project, providing further justification beyond the scope of traditional analysis.

Synergies

Backcasting began with a broad vision for Viet Nam rather than straightforward transport targets. An important reason for this is the identification of synergies between transport policy and Viet Nam's broader development programme.

Other infrastructure projects can piggyback on transport investment, reducing their combined cost. For example, the planned North-South Express Railway could facilitate the co-development of water and energy infrastructure.

More generally, transport corridors shape long-term growth, and policy can be designed strategically to foster growth and create jobs. Linking to the previous section on externalities, forecasting methods can be used to identify where synergies are greatest.

Implementation Framework

The structure of this brief broadly reflects the order of anticipatory analysis, from initial Backcasting through to granular econometric forecasting. However, it was noted that the Backcasting vision might need to be adjusted as Pathway Analysis clarifies the difficulties. Similarly, stress-testing of granular forecasting models might inform adjustments to the assumptions of both Backcasting and Pathway Analysis. The analysis process is therefore iterative.

Further, the relevance of an analysis erodes over time as new information emerges. The process of policy design and implementation should therefore be cyclical. Figure 2 proposes a development and reporting cycle to incorporate anticipatory analysis. The complete cycle should align with other government strategy reporting cycles, typically five years, to incorporate updated information in the Backcasting process.

Implementing transport policy and gathering data to inform new iterations of anticipatory analysis requires significant coordination. Hierarchy and Control theory informs a distribution from high-level oversight at the national level down to localised implementation. Figure 3 illustrates a simple framework from Tiwari et al. (2008) to distribute responsibility between stakeholders. The choice of Micro-level indicators is critical here. Detailed localised health and pollution data are needed for Qualitative Comparative Analysis, in particular, to inform the choice of future projects.

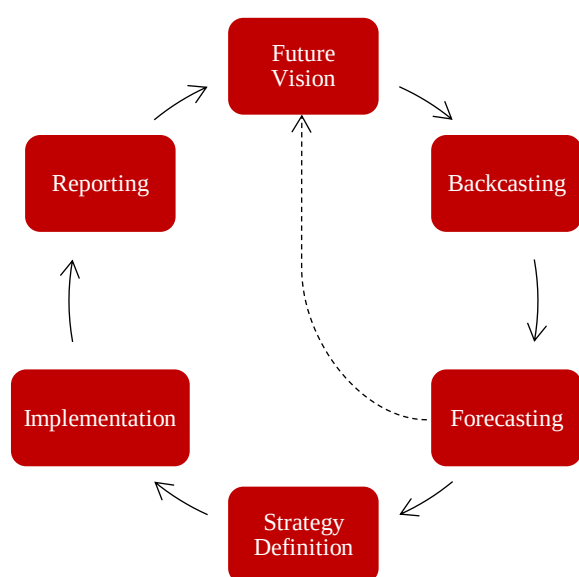


Figure 2: Five-Year Development and Reporting Cycle.

		QUALITATIVE		QUANTITATIVE	
SYSTEM HIERARCHY OF	THE CONTROL CYCLE				
	Objectives	Policies	Instruments	Indicators	
MACRO	LEVEL 1	Objective 1,1	Policy 1,1	Measure 1,1	Indicator 1,1
	...	...	...	...	
	Objective 1,n	Policy 1,n	Measure 1,n	Indicator 1,n	
	LEVEL 2	Objective 2,2	Policy 2,2	Measure 2,2	Indicator 2,2
	...	...	...	...	
	Objective 2,n	Policy 2,n	Measure 2,n	Indicator 2,n	
MICRO	LEVEL N	Objective n,n	Policy n,n	Measure n,n	Indicator n,n
	...	...	...	...	
	Objective n,n	Policy n,n	Measure n,n	Indicator n,n	

Figure 3: Hierarchy and Control Matrix.

Policy Recommendations

Implementing a policy development cycle based on anticipatory analysis is recommended for Viet Nam's transport sector. This will support health-conscious, ambitious, and strategic decision-making. The following implementation timeline is recommended:

Year One

Establish the Backcasting vision. Our National Green Growth Strategy is a useful starting point for the Backcasting process, but further collaboration across government departments is recommended. Collaboration across government departments will also create political buy-in, improving support for the strategy.

With the Backcasting vision established, the Ministry of Construction/Transport is recommended to scale up modelling and evaluation expertise. Given the importance of financial considerations, establishing econometric forecasting teams to quantify externalities and synergies should be prioritised.

Data infrastructure investments proposed in the National Action Plan on Pollution Remediation and Air Quality Management must be made in earnest. This is necessary to construct a Digital Twin of the transport system.

Year Two

Pilot Qualitative Comparative Analysis (QCAs) are recommended to evaluate flagship projects, for example, the Ho Chi Minh City subway, to build the investment case for future projects. This stage of implementation sequencing is important for drawing attention to successful strategic projects and garnering public and government support for more ambitious policy.

The Digital Twin should be improved over time to incorporate more information as the modelling team expands and data infrastructure improves. For example, modelling Disability-adjusted life years (DALY) as a function of vehicle fleet electrification would enrich the air quality analysis.

Year Three Onwards

Fully implement the policy development cycle proposed in Figure 2. A five-year cycle will provide sufficient time for comprehensive analysis and create a stable policy environment, while remaining responsive to new developments.

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